# L3: The G(n,p) model (ER Graphs)

Let's consider now the simplest random graph model and its degree distribution.

This model is referred to as G(n, p) and it can be described as follows: the network has n nodes and the probability that any two distinct nodes are connected with an undirected edge is p.

The model is also referred to as the Gilbert model, or sometimes the Erdős–Rényi (ER) model, from the last names of the mathematicians that first studied its properties in the late 1950s.

Note that the number of edges m in the G(n,p) is a random variable. The expected number of edges is $p \cdot \frac{n(n-1)}{2}$, the average node degree is $p \cdot (n-1)$ , the density of the network is $p$ and the degree variance is $p \cdot (1-p) \cdot (n-1)$ . These formulas assume that we do not allow self-edges.

The degree distribution of the G(n,p) model follows the Binomial(n-1,p) distribution because each node can be connected to n-1 other nodes with probability p.

Note that the G(n,p) model does not necessarily create a connected network – we will return to this point a bit later.

Also, in the G(n,p) model there are no correlations between the degrees of neighboring nodes. So, if we return to the friendship paradox, the average neighbor degree at a G(n,p) network is $\overline{k_{nn}} = \bar{k} + (1-p)$ (using the Binomial distribution) -- or $\overline{k_{nn}} = \bar{k} + 1$ (using the Poisson approximation when $p \ll 1$) .

In other words, if we reach a node v by following an edge from another node, the expected value of *v*'s degree is one more than the average node degree.

***Food-for-thought:*** Derive the previous expressions for the average neighbor degree with both the Binomial and Poisson degree distributions.